

Exam in “Economics & Law”

Winter Semester 2024/2025

Please read carefully the following instructions:

- This is a written exam based on 3 questions.
 - Each answer awards a different amount of points, as stated in brackets at the end of the question. The total points sum up to 100.
- This is an open-book exam. You may bring with you any written/printed material but you are not allowed to use a computer.

Questions:

Q1. Suppose there is a market for robots that is characterized by regular supply and demand curves. Three events occur one after the other: First, the government imposes a per-unit tax on the production of robots. Second, there is an increase in the demand for AI-processors, which are used as an input in the production of robots. Third, there is a decrease in the income of all consumers.

(a) Explain how this chain of events is expected to effect the equilibrium quantity, equilibrium price, and efficiency in the market for robots. [15 points]

Per-unit tax reduces supply (+ causes a deadweight loss). Increase in demand for the (complement) input increases its price, which further reduces the supply of robots through higher marginal costs. The decrease in income generally causes demand to decrease (unless it's inferior goods, which seems unlikely for robots). Hence: quantity decreases (less supply, less demand), but effect on price is ambiguous. Efficiency decreases because of the DWL created by the tax.

(b) The government is considering imposing a lump-sum tax instead of a per-unit tax. If this occurred, how would this affect your answer to section (a)? [5 points]

On the one hand, lump-sum taxes don't directly affect production and hence don't result in direct DWL. On the other hand, it may cause firms to exit in the long term, reducing supply still.

(c) Suppose that Germany can produce either 1 Robot or 2 Cars in any given minute, but Canada can produce 3 Robots and 4 Cars at any given minute. Who should produce which product, and why? [10 points]

Comparative advantages (mark the cheaper option in bold):

	Robot	Car
Germany	1	2
Canada	3	4

	Cost of producing 1 Robot	Cost of producing a Car
Germany	2 cars	½ robot
Canada	4/3 cars	¾ robot

Germany should produce cars, whereas Canada should produce robots.

Q2. Adam and Ben are two friends who decide to set up a startup together. They need to decide who will be the CEO and who will be the COO and they agree on the following procedure: each person privately writes his name on a note and states “CEO” or “COO”. Then, they read the notes together. If one person write “CEO” whereas the other writes “COO”, then the CEO gets a yearly salary of 6 Million Euros, whereas the COO gets a salary of 2 Million Euros. If both write “COO” then the startup never gets formed, and they each get zero. If both write “CEO”, they become co-managers and each earn a salary of 4 Million Euros.

(a) Construct a payoff table. [7 points]

	Ben: CEO	Ben: COO
Adam: CEO	4, 4	6, 2
Adam: COO	2, 6	0, 0

(b) State for each of the cells in the table whether it is Kaldor-Hicks Efficient and whether it is Pareto Efficient. [8 points]

All three cells with positive payoffs are KH efficient and therefore, by definition, Pareto efficient. The cell with (0,0) is neither KH efficient nor Pareto efficient.

(c) What is the Nash Equilibrium (or Nash Equilibria if there is more than one) of this game? [5 points]

Both parties have a dominant strategy of choosing “CEO”. Hence, There is only one NE – (CEO, CEO) yielding (4,4).

(d) Suppose that the parties could bargain at zero transaction costs. How would this affect the outcome of the game? [5 points]

It wouldn't, there is nothing to be gained from trade here.

(e) Suppose that the negotiation would instead work as follows: Adam announces aloud first whether he wants to be CEO or COO. Then, Ben responds after hearing the declaration. The payoffs otherwise remain the same. How would this change affect the outcome of the game? [5 points]

It wouldn't, because both have a dominant strategy, so converting the game into a dynamic one will not affect anything.

(f) Suppose that Adam and Ben would only earn 1 Million as co-managers. How would this affect your analysis in the previous subsections? [15 points]

	Ben: CEO	Ben: COO
Adam: CEO	1, 1	6, 2
Adam: COO	2, 6	0, 0

We now have 2 NE (CEO, COO) and (COO, CEO) yielding (6,2) and (2,6). Now (CEO, CEO) is no longer KH efficient or Pareto efficient. If they can negotiate at zero TC, still nothing would change, because (1,1) is not an equilibrium. If Adam moves first, we get that only (6,2) is a Subgame perfect NE.

Q4. The “Republic of Nimbala” (a hypothetical country) is a resource-rich developing nation that emerged from a civil war three years ago. Despite its mineral wealth and agricultural potential, the country faces several challenges. The hastily negotiated peace agreement created a fragmented government where former war factions control different ministries. The Minister of Mining, a former warlord, requires all foreign companies to partner with local firms owned by his family members before receiving licenses. Only 30% of declared mining revenues reach the national treasure, despite some collection efforts. Furthermore, land ownership is contested between colonial-era title holders, indigenous communities claiming ancestral rights, and war refugees resettled by various government programs. Court backlogs for land disputes exceed five years. Transactions in movable assets (including cars, furniture, and others) are regulated by the housing commission, led by the president’s daughter, who is insisting that all transactions must be documented by carving the names of the parties into a wooden desk in her office. Moreover, business registration requires navigating 15 different government agencies, with each step demanding unofficial “speed processing fees.” Many officials maintain luxury properties abroad despite modest salaries. A recent microfinance program also revealed that poor people are unable to get access to credit, because the government’s “directed lending program” prohibits giving loans to anyone who cannot prove they have assets worth at least 5,000 Euros.

(a) Identify at least six distinct institutional and economic problems that Nimbala is facing. For each problem, explain the underlying causes and how it impedes economic development. **[16 points]**

Sample solution [very briefly; full answers are expected to get full points]:

- (1) There is no longer war, but the division to factions still causes war-like instabilities, which would discourage foreign investments.
- (2) the requirement for partnering with local firms can discourage foreign partners, if local firms are inefficient. The fact that it’s firms owned by family members imply corruption / rent seeking.
- (3) tax base seems ineffective (only 30%)
- (4) slow courts cause property rights to be ineffective
- (5) the registration of movables is odd – is it necessary? The method of engraving into a table increases TC.
- (6) inability to get access to credit – seems related to the registration problems of property rights
- (7) directed lending being inefficient

(b) Propose a specific solution for three of the problems you identified in Part A. At least one solution must apply concepts from behavioral law and economics. For each solution, explain how it addresses the underlying causes and discuss potential implementation challenges **[9 points]**

Sample solution [very briefly; full answers are expected to get full points]:

- 1) Establish a clear property rights registry, which will lower TC and increase certainty. This might be tough to do given the existing state of registration and may invoke objections from the existing government.
- 2) Streamline business registration – one stop shop, get rid of fees. Challenge: the fees may be needed to run the system, one-stop-shop may be expensive, etc.
- 3) Improve the speed of courts / eliminate registration of movables. Challenges: speed comes at the expense of quality, convincing the president’s daughter.

As a behavioral concept: use nudges (e.g. change the default, use framing, send a personalized letter to increase tax collection, etc.)